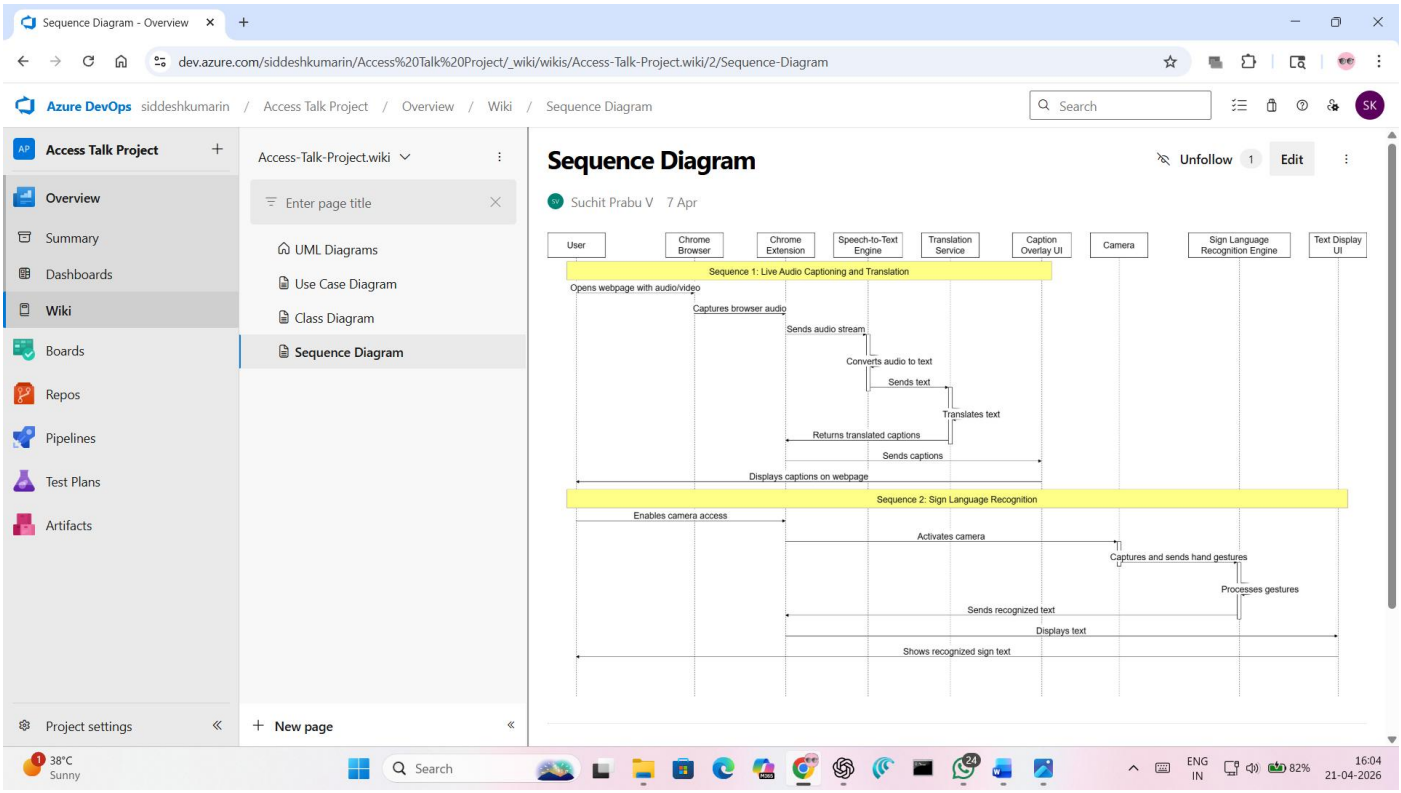
Experiment 5 : Sequence Diagram

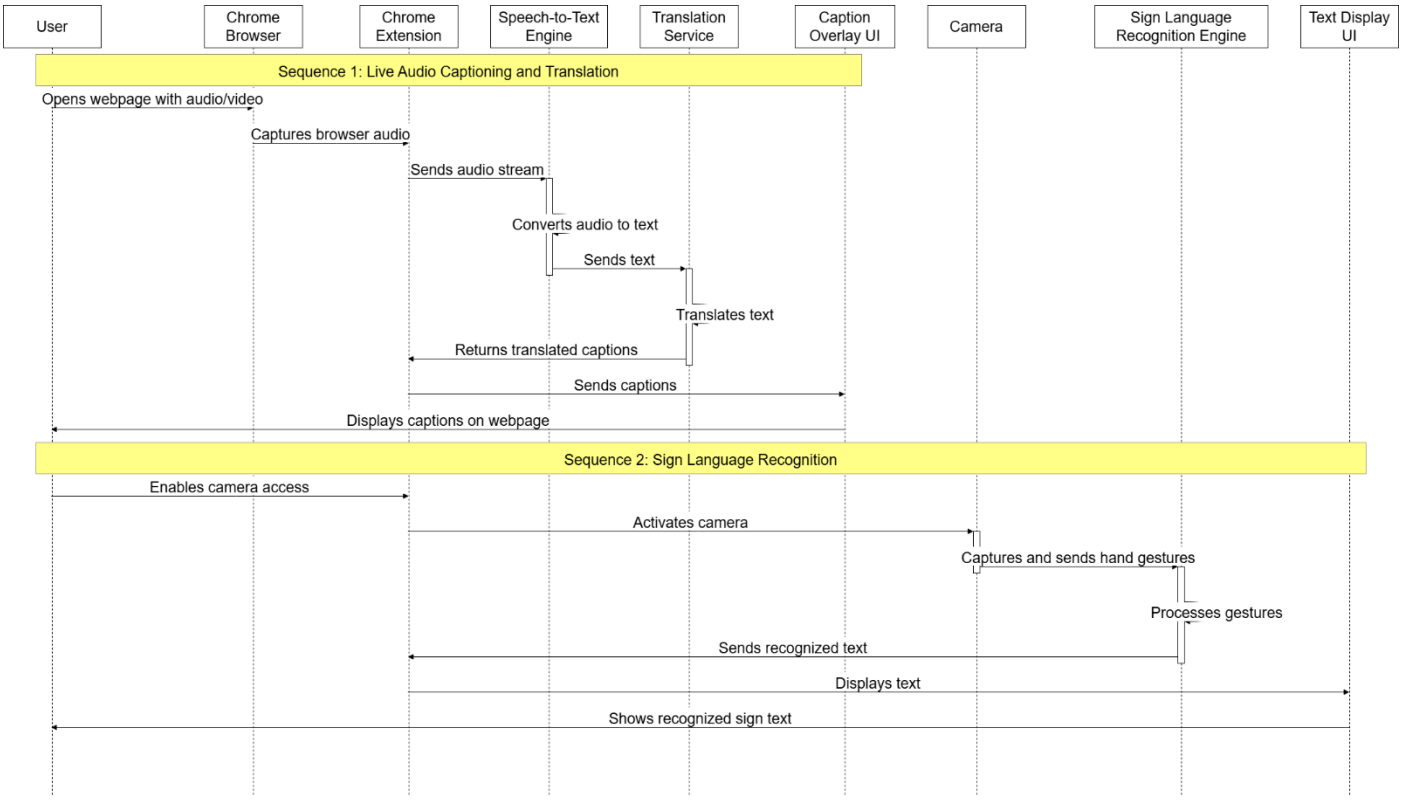
Steps to create diagram

1. Open draw.io (diagrams.net)
2. Create Blank Diagram
3. Add lifelines (vertical dashed lines) for:
   User, Chrome Browser, Chrome Extension, Speech-to-Text Engine, Translation Service, Caption Overlay UI, Camera, Sign Language Recognition Engine, Text Display UI
4. Add two sections (use rectangle with light color):
   a. Sequence 1: Live Audio Captioning and Translation
   b. Sequence 2: Sign Language Recognition
5. Connect arrows:
   a. Sequence 1 (Audio flow):
   b. Sequence 2 (Sign flow):
   User → Chrome Extension → Camera → Sign Language Recognition Engine → Text Display UI
6. Arrange layout:
   a. Left = Audio captioning flow
   b. Right = Sign recognition flow
7. Export as PNG

Diagram Notations

1. User → initiates actions (audio or sign)
   - Chrome Browser → source of webpage audio
   - Chrome Extension → controls and connects all components
2. Speech-to-Text Engine → converts audio to text
   - Translation Service → translates text into captions
   - Caption Overlay UI → displays captions on screen
3. Camera → captures hand gestures
   - Sign Language Recognition Engine → processes gestures into text
   - Text Display UI → shows recognized sign text


Sequence Diagram - Overview
dev.azure.com/siddeshkumarin/Access%20Talk%20Project/_wiki/wikis/Access-Talk-Project.wiki/2/Sequence-Diagram
Azure DevOps siddeshkumarin / Access Talk Project / Overview / Wiki / Sequence Diagram
Access Talk Project
Overview
Summary
Dashboards
Wiki
Boards
Repos
Pipelines
Test Plans
Artifacts
Project settings
Access-Talk-Project.wiki
Enter page title
UML Diagrams
Use Case Diagram
Class Diagram
Sequence Diagram
New page
Sequence Diagram
Unfollow
Edit
Suchit Prabu V 7 Apr
User
Chrome Browser
Chrome Extension
Speech-to-Text Engine
Translation Service
Caption Overlay UI
Camera
Sign Language Recognition Engine
Text Display UI
Sequence 1: Live Audio Captioning and Translation
Opens webpage with audio/video
Captures browser audio
Sends audio stream
Converts audio to text
Sends text
Translates text
Returns translated captions
Sends captions
Displays captions on webpage
Sequence 2: Sign Language Recognition
Enables camera access
Activates camera
Captures and sends hand gestures
Processes gestures
Sends recognized text
Displays text
Shows recognized sign text


User
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Speech-to-Text Engine
Translation Service
Caption Overlay UI
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